

Dark Matter Deficient Galaxies (DMDGs) And Their Member Star Clusters (SCs) Form Simultaneously during High-velocity Galaxy Collisions in 1.25 pc Resolution Simulations

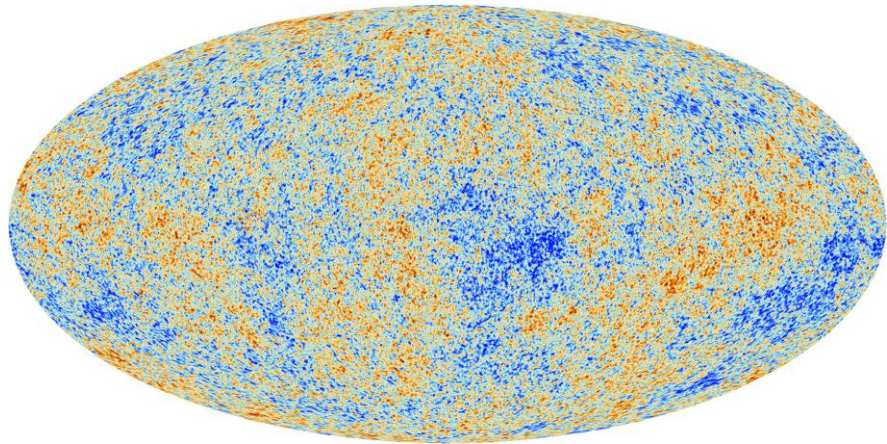
Joohyun Lee, Eun-jin Shin, Ji-hoon Kim



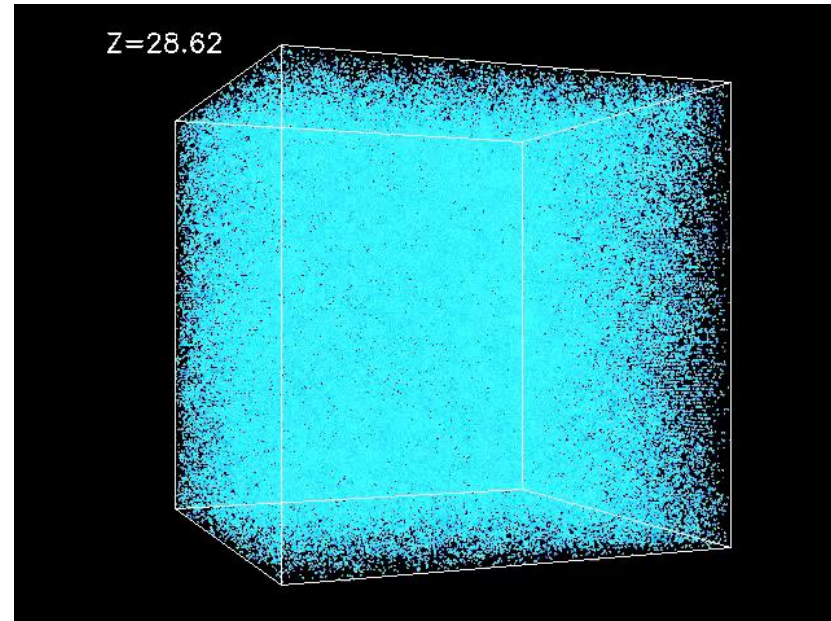
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AGORA Workshop 2021

Cosmological Structure Formation in Λ CDM Cosmology – Dark Matter Halo Formation

- In Lambda-CDM (Λ CDM) cosmology paradigm, structure formation occurs in a “bottom-up” manner.
- Starting from primordial density fluctuation, dark matter (DM) gravitationally collapses to form DM halos.
- Small DM halos form firstly and merge, producing more massive halos.



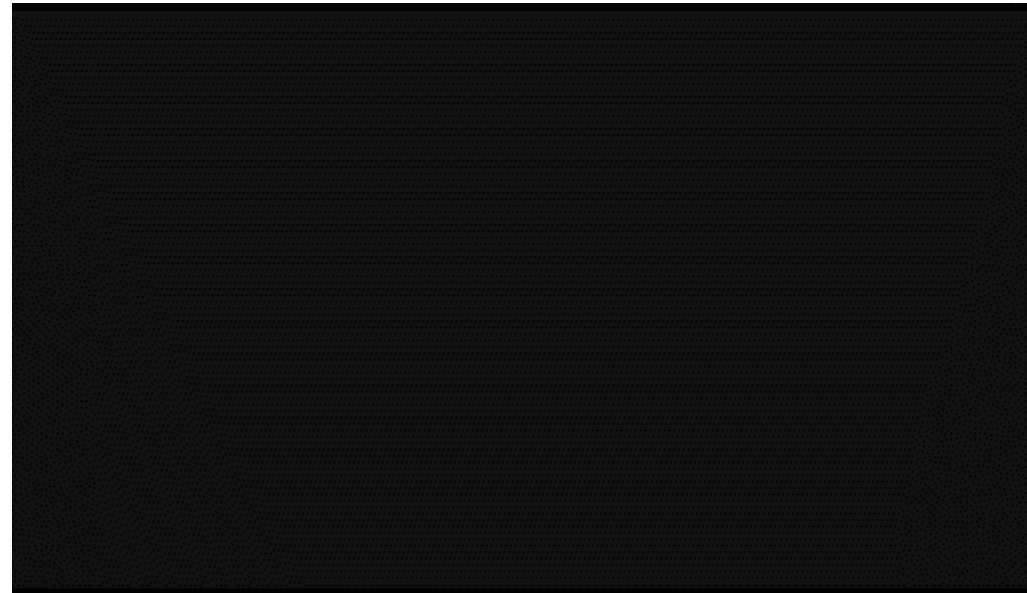
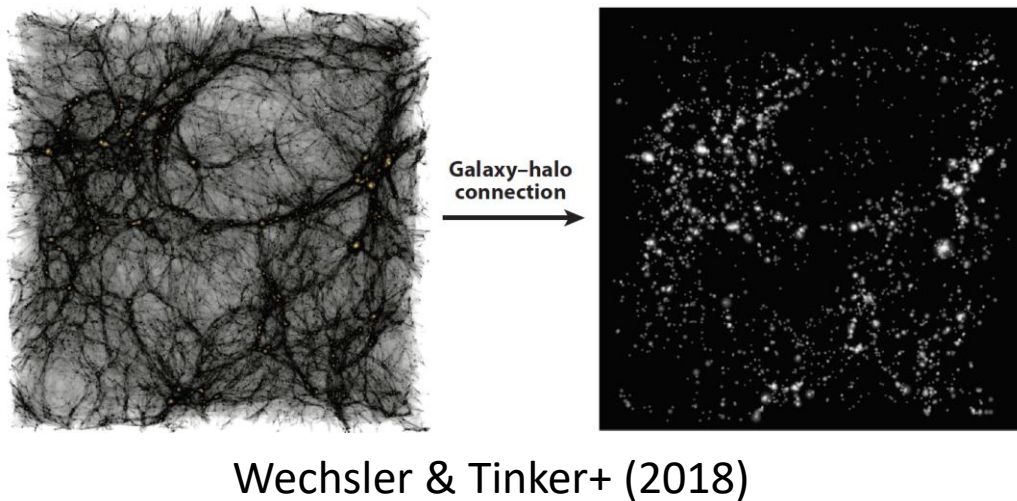
Primordial density fluctuation from CMB
Credit: Planck Collaboration (2018)



Clustering of DM
Credit: Andrey Kravtsov

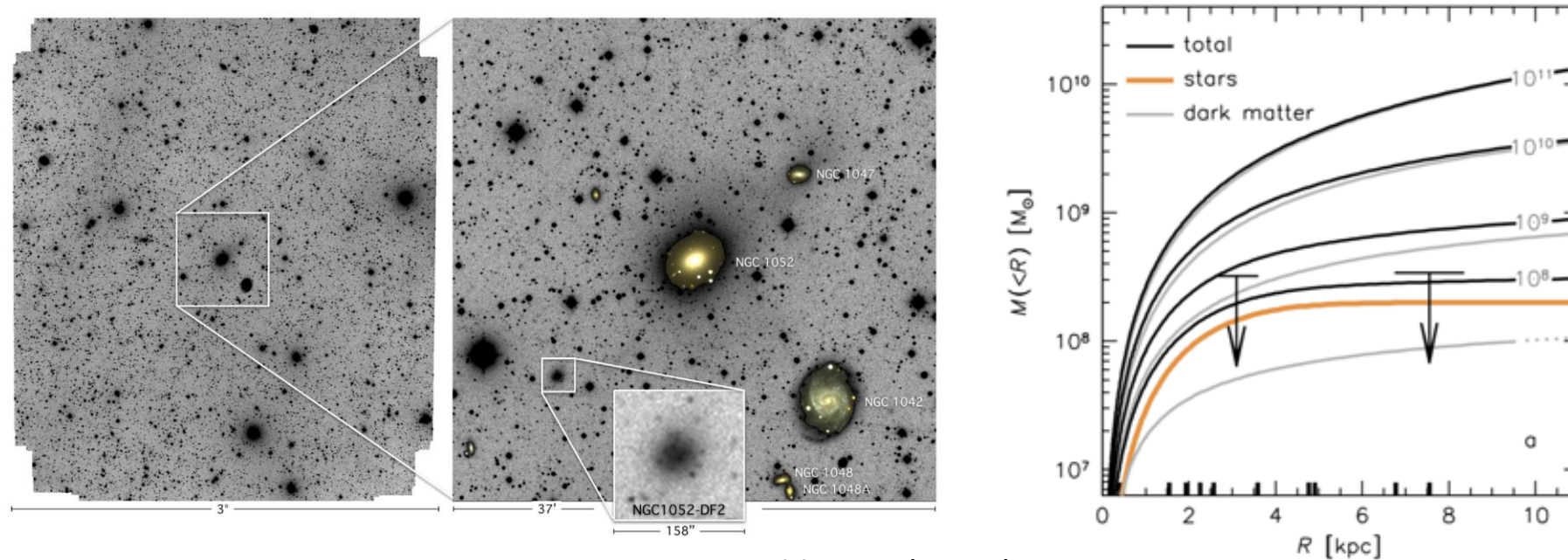
Cosmological Structure Formation in LCDM Cosmology – Galaxy and Galaxy Cluster

- Since baryons have high temperature, they need more time to cool.
 - DM firstly builds the skeleton of baryonic structures like galaxies and galaxy clusters.
 - Then, gas fall into sufficiently massive DM halo and forms stars and galaxies.
- **Most galaxies contain a great amount of dark matter.**



Infall of gas to halos
Credit:
IllustrisTNG project

Discovery of Dark Matter Deficient Galaxies (DMDGs) – NGC1052-DF2 & NGC1052-DF4

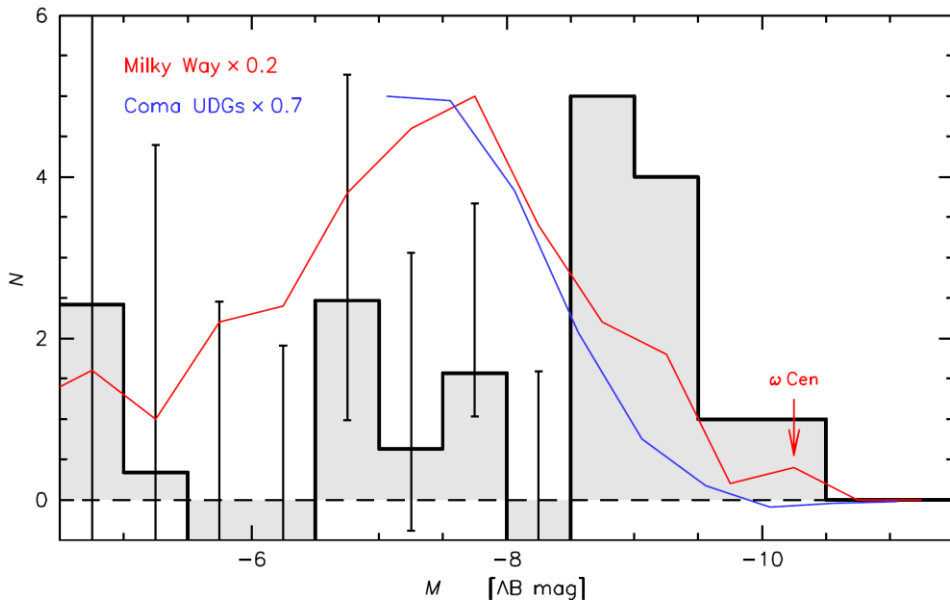


van Dokkum+ (2018)

- NGC1052-DF2 (DF2) and NGC1052-DF4 (DF4), Ultra Diffuse Galaxies (UDGs) with $M_{\text{star}} \sim 2 \times 10^8 M_{\odot}$ are found to have only little DM (van Dokkum et al. 2018 & 2019).
- The mysterious small fraction of DM in those dwarf galaxies ***cannot be explained*** by the conventional galaxy formation theory in the LCDM cosmology.

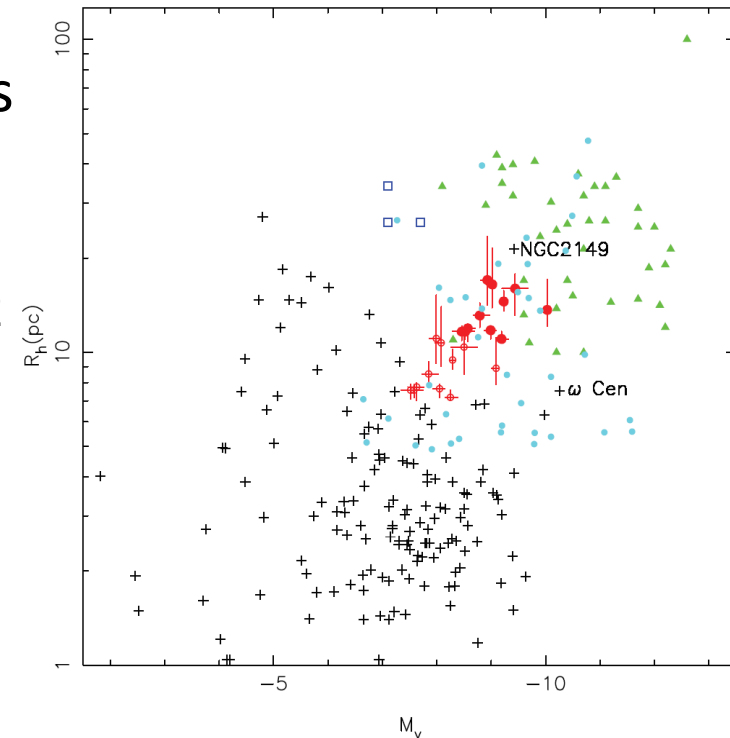
Extraordinary Properties of Globular Clusters (GCs) in NGC1052-DF2

- NGC1052-DF2 has an unconventionally bright population of globular clusters (GCs) (van Dokkum et al. 2018, Ma et al. 2020, Shen et al. 2021).
- GC population has a peak at $M_V \sim -9$ mag, which is ~ 1.5 mag brighter than mean luminosity of typical Milky Way GCs.
- Their size and ellipticities are also very large compared to MW GCs



van Dokkum+ (2018)

●: GCs of DF2
Ma+ (2020)



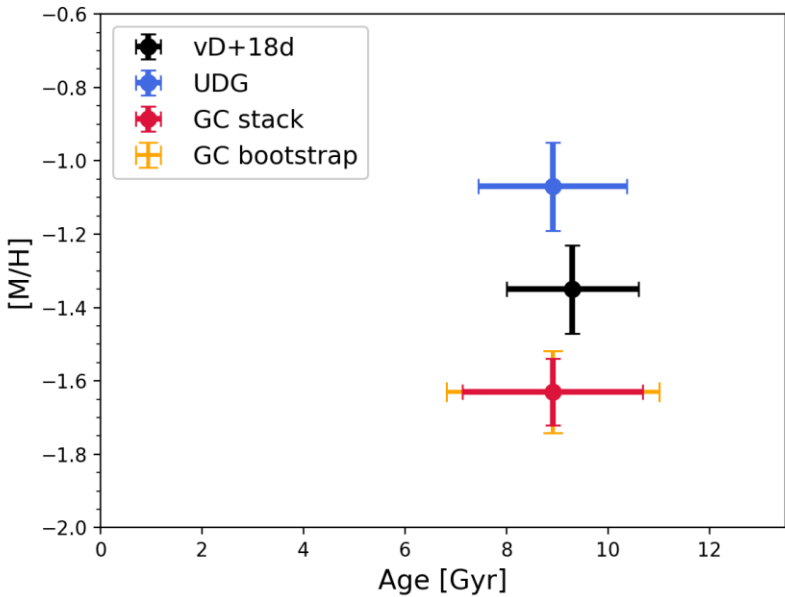
Other Observed Properties of NGC1052-DF2

- The age of NGC1052-DF2 is $\sim 8\text{--}10$ Gyr (Ruiz-Lara 2019, Fensch 2019).
- The metallicity of NGC1052-DF2 stellar body is $[M/H] \sim -1.0 \text{--} -1.2$ (Ruiz-Lara 2019, Fensch 2019).
- Fensch et al. (2019) observed small difference between metallicity of GCs and the stellar body.

Region	S/N (pixel^{-1})	V (km/s)	LW Age (MW) (Gyr)	LW $[M/H]$ (MW) (dex)
(1)	(2)	(3)	(4)	(5)
$R < 1/4R_e$	43	1845 ± 46	8.7 ± 0.7 (9.8 ± 0.5)	-1.18 ± 0.05 (-1.20 ± 0.07)
$1/4R_e < R < 1/2R_e$	21	1784 ± 55	8.6 ± 1.0 (9.8 ± 0.9)	-1.1 ± 0.1 (-1.2 ± 0.1)

Table 2. Parameters derived for the two analysed regions. (3) Heliocentric-corrected velocity computed using pPXF; (4) and (5) average age and metallicity estimations (both, light and mass-weighted) from the SFH recovered combining all STECKMAP tests.

Ruiz-Lara+ (2019)

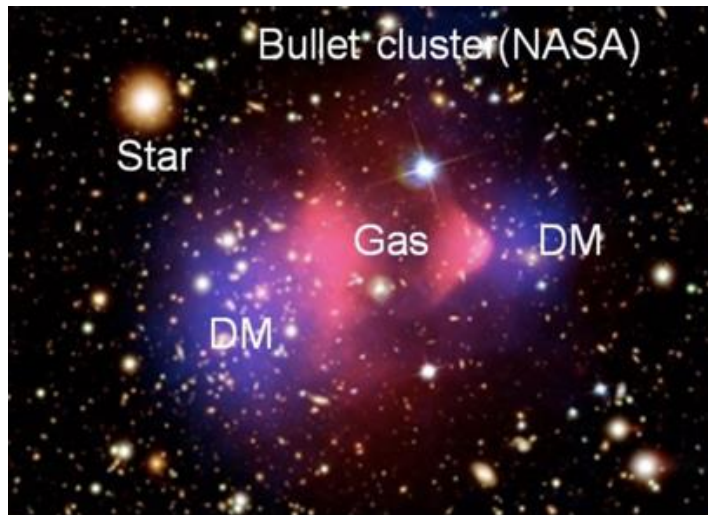


Fensch+ (2019)

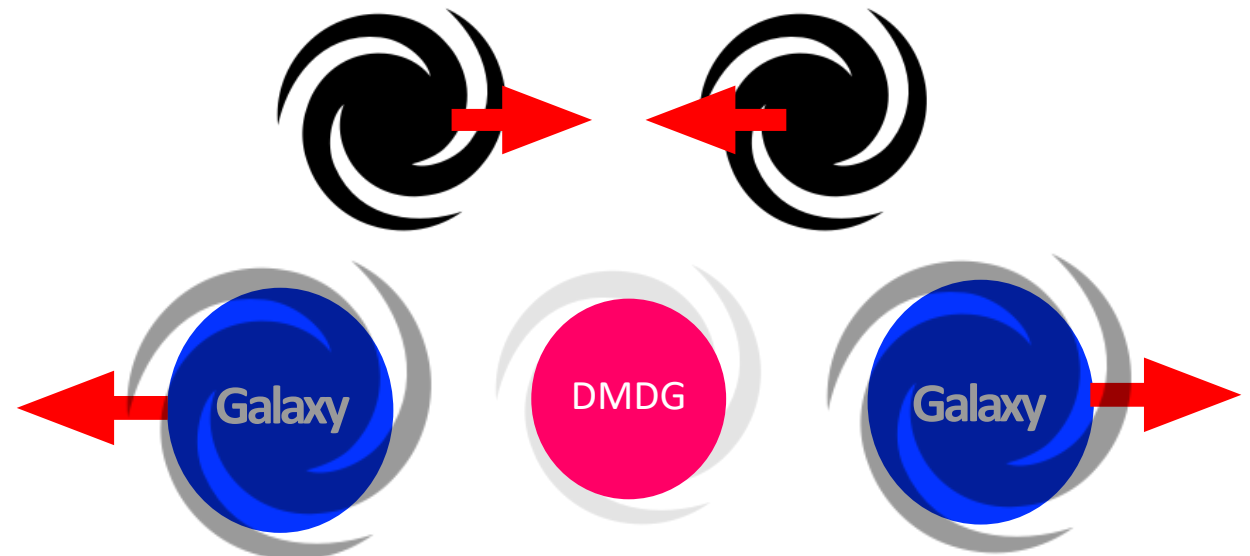
DM-Deficient Galaxies Can Form during High-velocity Galaxy Collision

- Silk (2019) suggested that **high-velocity** collision of **gas-rich dwarf galaxies** can produce DMDGs.
- This is similar the famous "Bullet Cluster",
- Shock-compressed gas in the collision can form stars efficiently, invoking simultaneous formation of massive GCs and DMDG.

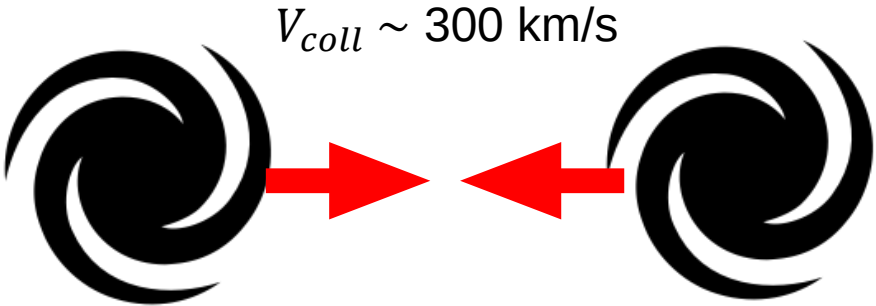
Cluster scale



Galaxy scale



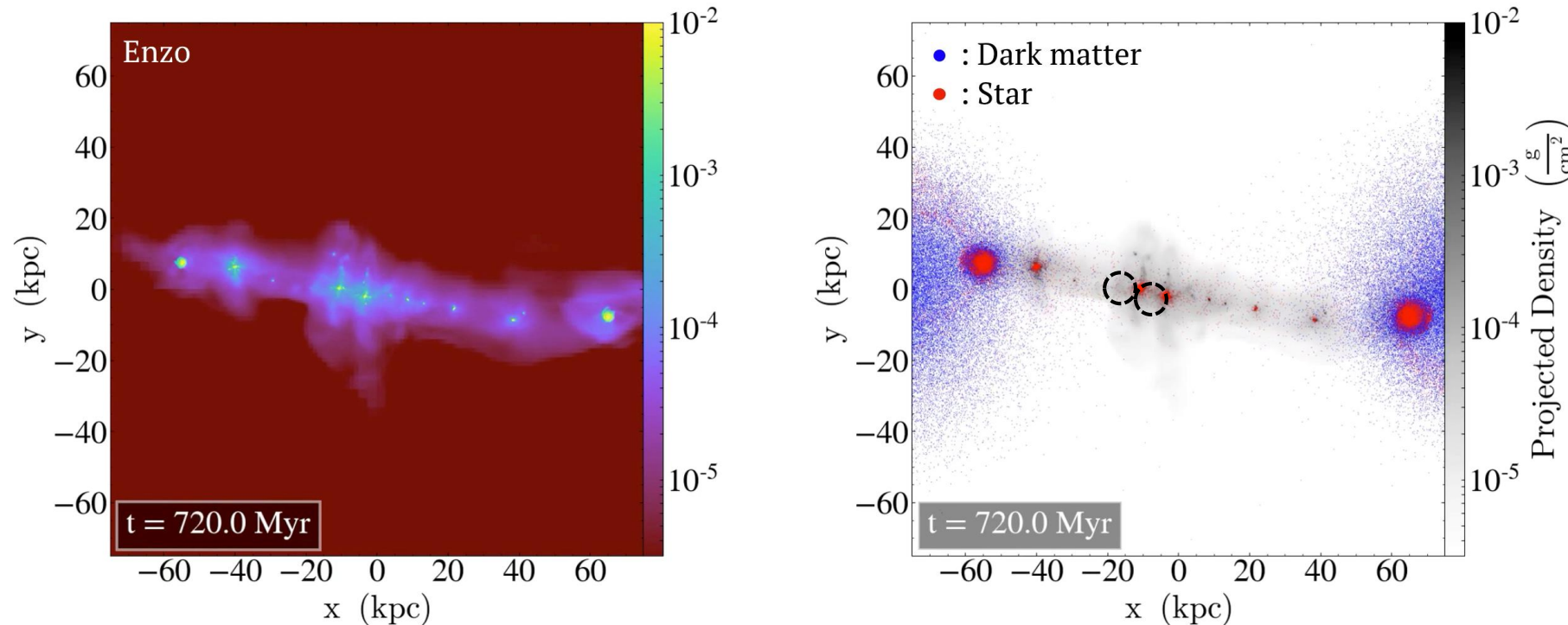
Simulation Setup—Initial Condition of Fiducial Run and Numerical Resolution



$$\begin{aligned} M_{\text{star}} &= 4.21 \times 10^8 M_{\odot} \\ M_{\text{gas}} &= 1.68 \times 10^9 M_{\odot} \\ M_{\text{DM}} &= 3.88 \times 10^9 M_{\odot} \end{aligned}$$

Run Name	Initial Metallicity ($Z_{\odot}$)	Stellar Feedback Energy ($10^{51} \text{ erg} / 150 M_{\odot}$)
Fiducial	1	1
FB3	1	3.3
FB7	1	6.7
FB10	1	10
Low-metal	0.01	1

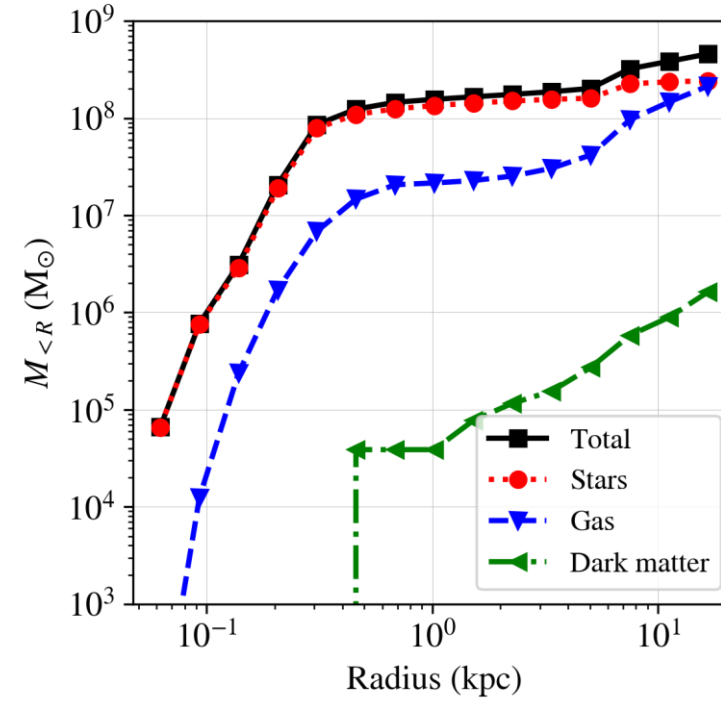
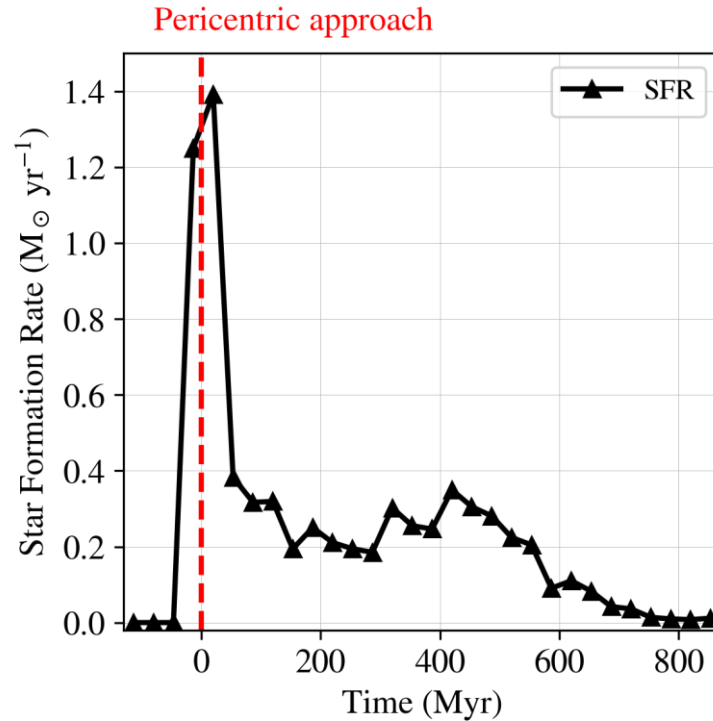
80 pc Resolution Simulation: High-velocity Collision Between Two Dwarf Galaxies



Shin+ 2020

- In our previous study Shin et al. (2020), we demonstrated that DMDGs can be produced during high-velocity collision of dwarf galaxies.
- Simulation is run for 850 Myr after the pericentric approach ($t = 150$ Myr).

80 pc Resolution Simulation: Properties of Produced DM-Deficient Galaxies (DMDGs)



Shin+ 2020

- The collision dramatically boosts the SFR to a maximum of $1.4 M_{\odot}/\text{yr}$.
- The most massive DMDG

$$M_{\text{star}} \sim 1.60 \times 10^8 M_{\odot}, M_{\text{gas}} \sim 4.17 \times 10^7 M_{\odot}, M_{\text{DM}} \sim 2.17 \times 10^6 M_{\odot}$$

$\Delta x = 80 \text{ pc}$ and $M_{star\ particle} = 40000 \text{ M}_{\odot}$ are not enough for resolving star formation of individual gas clouds.

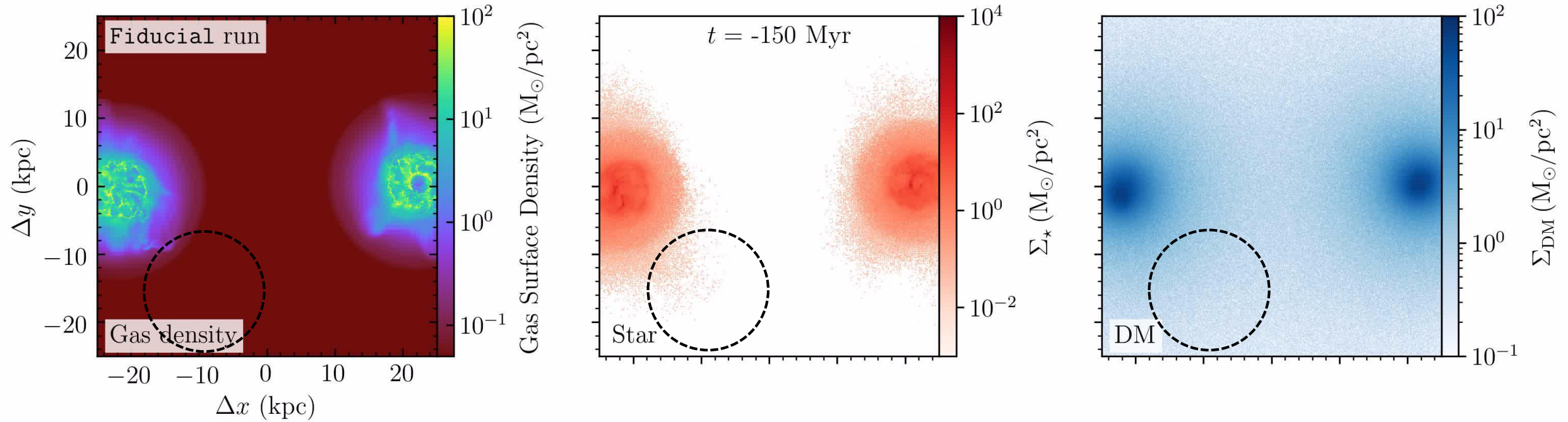


$\Delta x = 1.25 \text{ pc}$ and $M_{star\ particle} = 400 \text{ M}_{\odot}$ can resolve star formation in star cluster scale.

Improved Numerical Recipe and Subgrid Physics

	Previous work (Shin et al. 2020)	This work (Lee et al. 2021)
$N_{particle}$ (DM, stars)	$\sim 10^5$	$\sim 10^6$
Resolution (Δx)	80 pc	1.25 pc
$M_{star\ particle}$	40000 $M_\odot$	400 $M_\odot$
Refinement method	Simple adaptive mesh refinement (AMR)	AMR in nested refine region
Star formation threshold density (n_{th})	10 cm^{-3}	$4 \times 10^4\text{ cm}^{-3}$
Stellar feedback	Instantaneous thermal feedback	Gradual thermal feedback

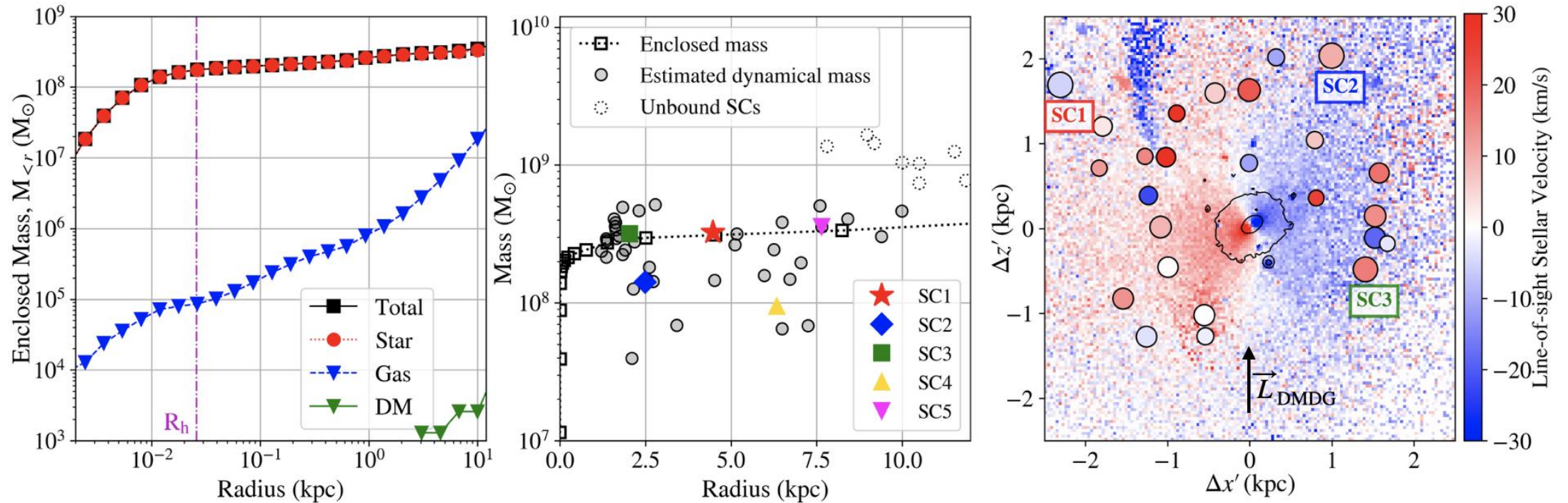
1.25 pc Resolution Simulation: High-velocity Collision Between Two Dwarf Galaxies



- The most massive DMDG has

$$M_{\text{star}} = 3.47 \times 10^8 M_{\odot} \text{ with almost no DM.}$$

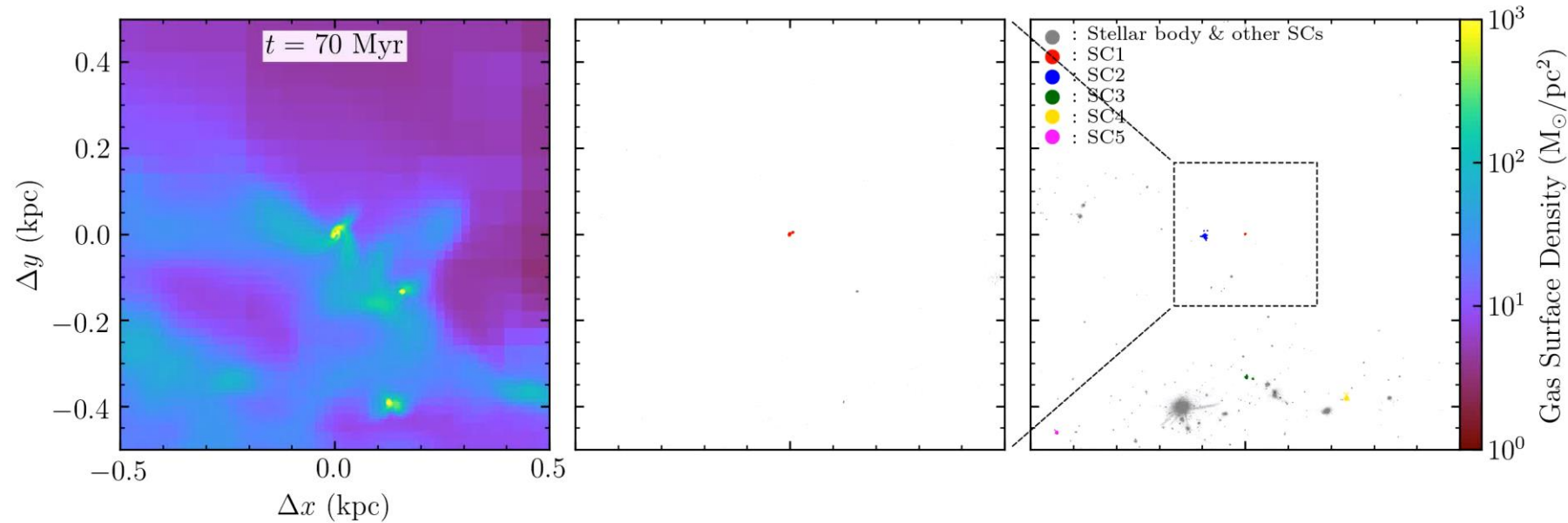
Formation of Star Clusters and DM-Deficient Galaxy (DMDG)



- 800 Myr after the pericentric approach, 42 gravitationally bound SCs orbit DMDG.
- SCs are being virialized, a key assumption of dynamical mass estimation (virial theorem).

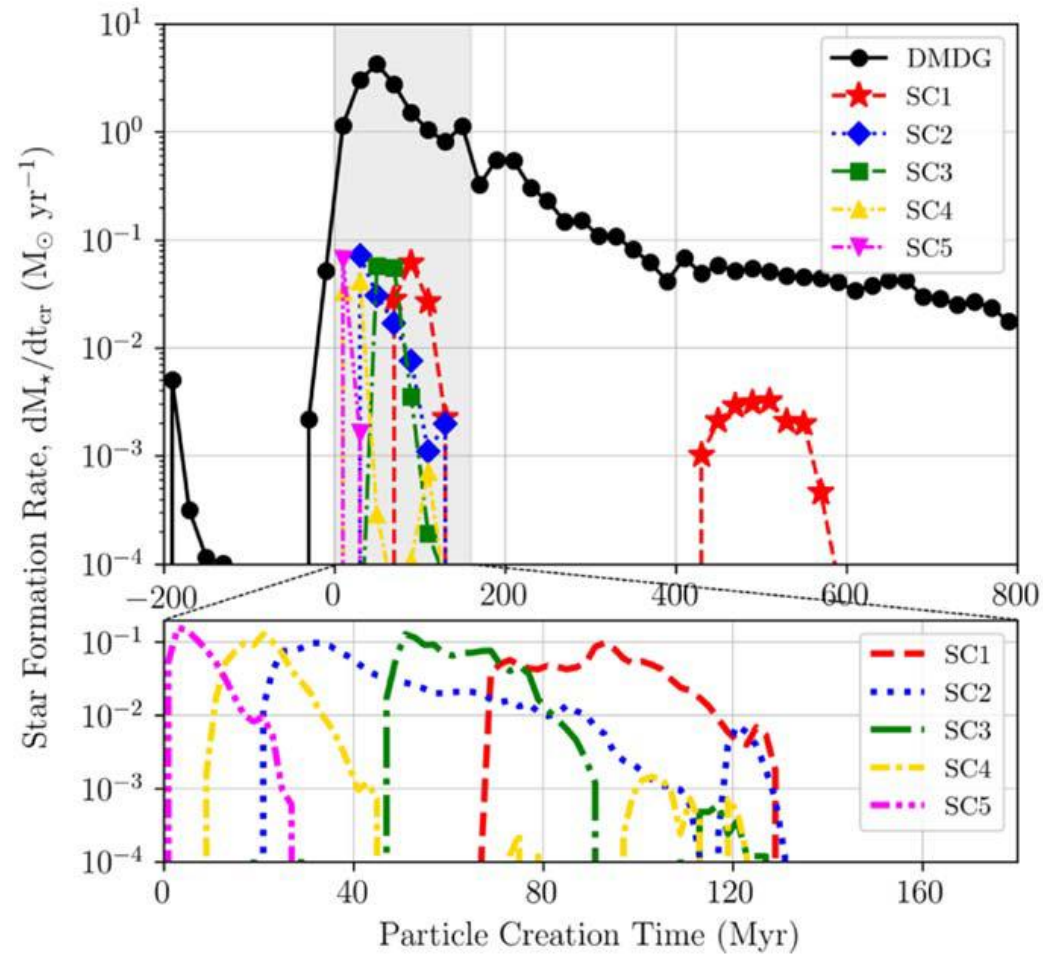
Relaxation timescale $t_{relax} \sim \text{few Gyr}$

1.25 pc Resolution Simulation: Resolving Star Cluster Formation



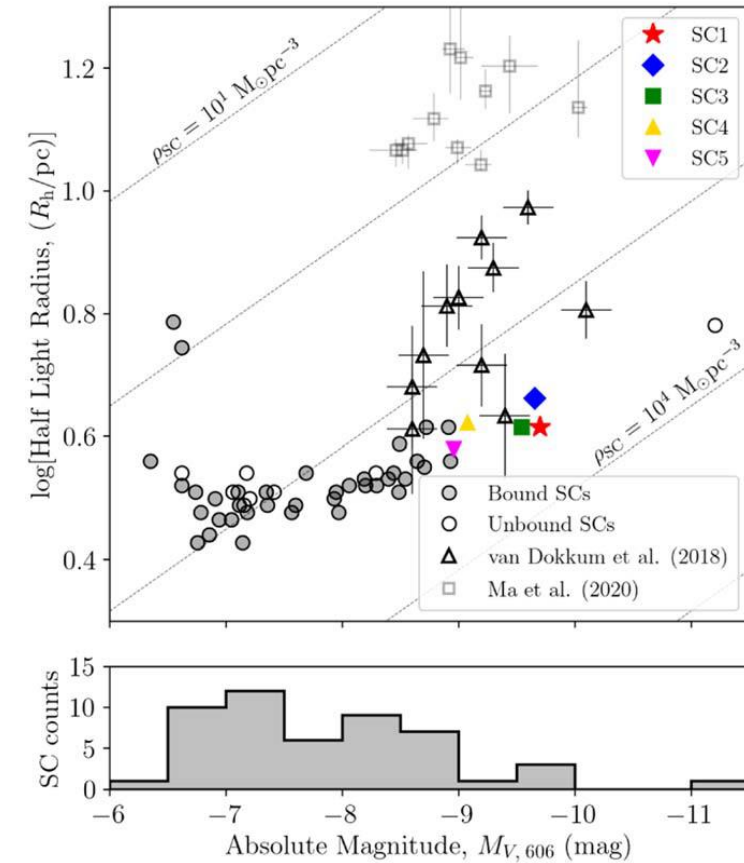
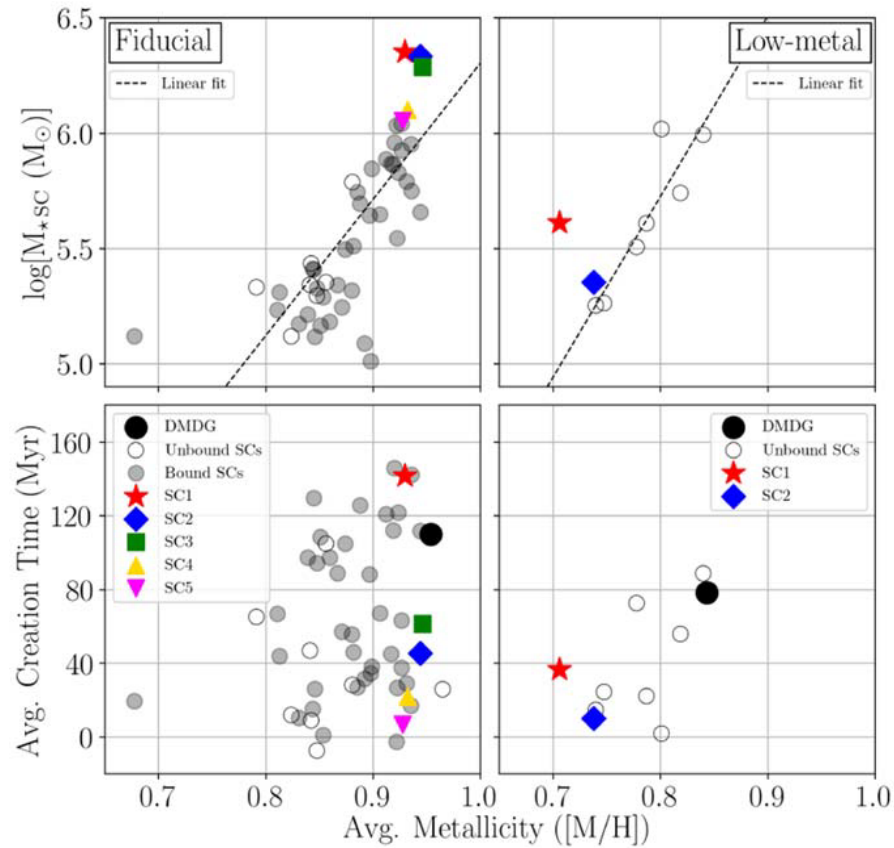
- SC (red) is formed in dense gas clump and orbits DMDG (gray).
 - Surface density in dense gas clumps: $\Sigma \geq 10^3 M_{\odot} \text{ pc}^{-2}$
- Efficient star formation, massive SC formation.

Simultaneous Formation of Massive Star Clusters and DM-Deficient Galaxy (DMDG)



- DMDG stars and SCs are formed simultaneously.
- Massive SCs are formed within 150 Myr after the pericentric approach.

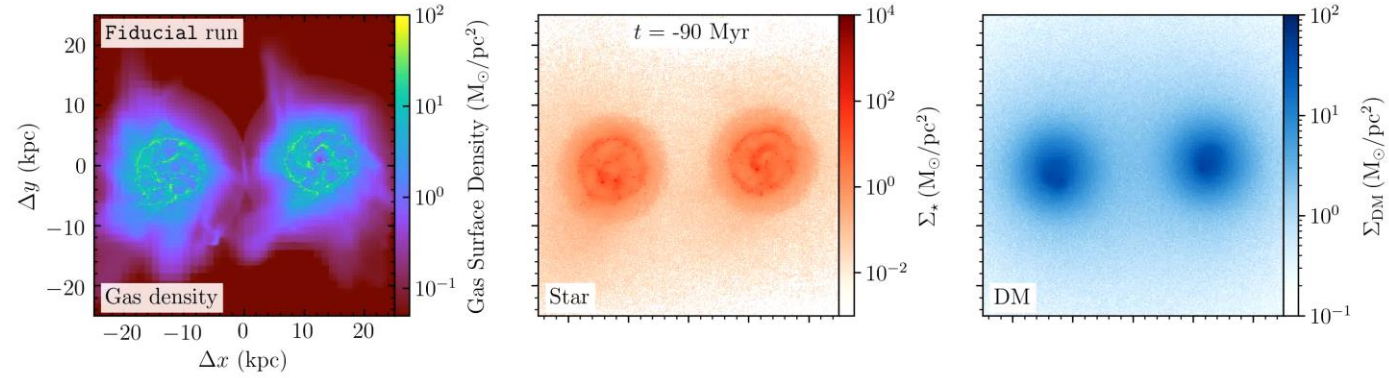
Physical Properties of Star Clusters (SCs)



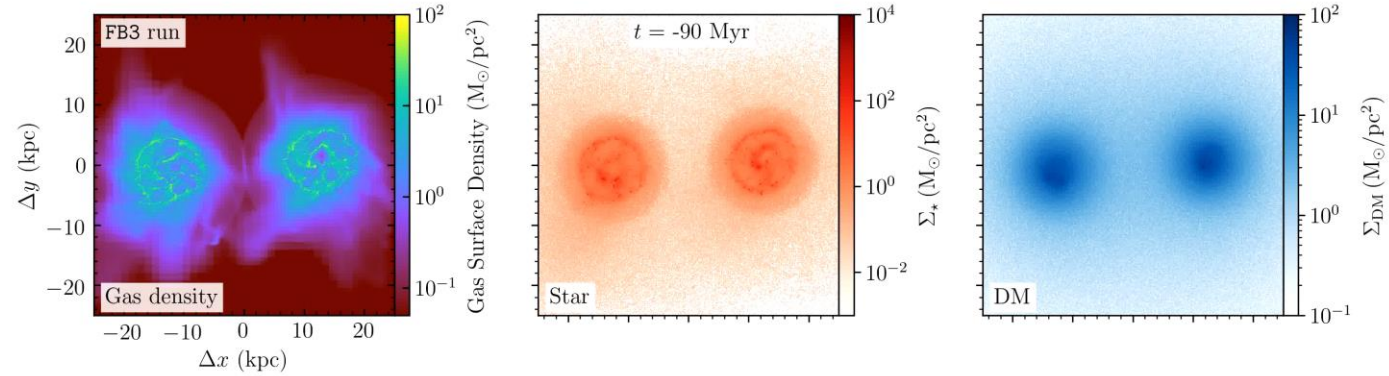
- Less number of SCs are created in Low-metal run ($Z = 0.01 Z_{\odot}$).
- SC population is slightly brighter than MW GCs and massive ($M_{\text{star, SC}} \sim 10^5 - 10^6 M_{\odot}$) SCs overlaps with the observation.

Parameter Study: Strength of Stellar Feedback

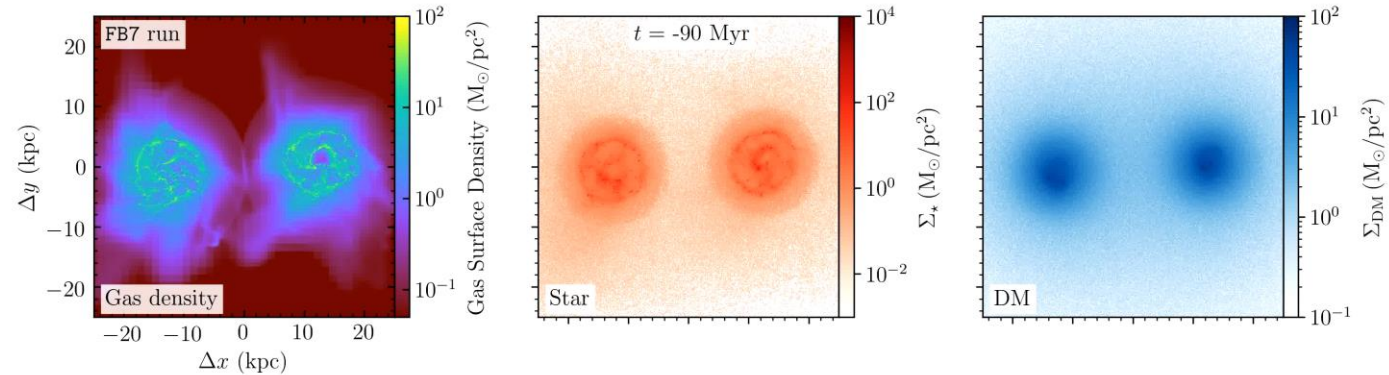
Fiducial run



$\times 3$ feedback energy

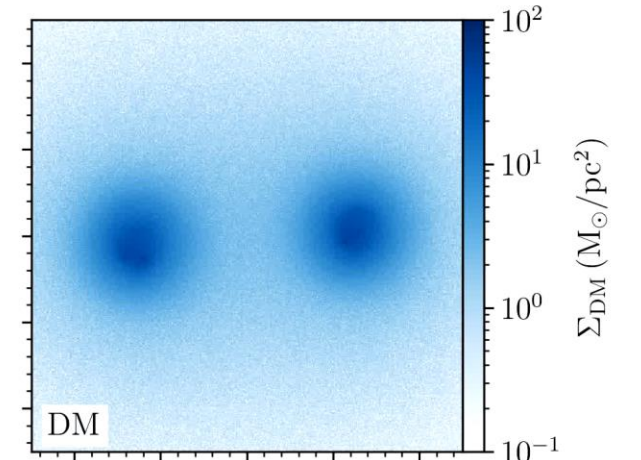
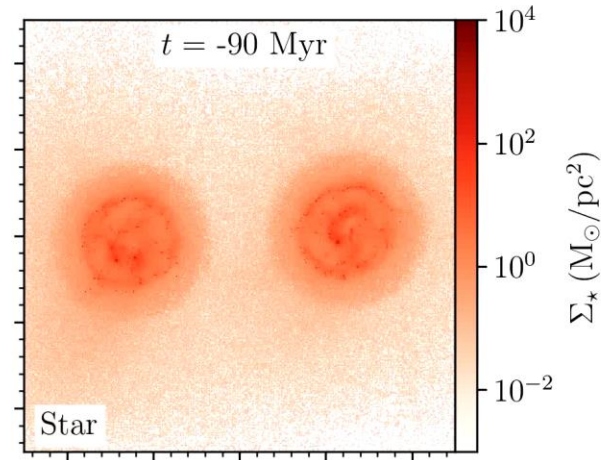
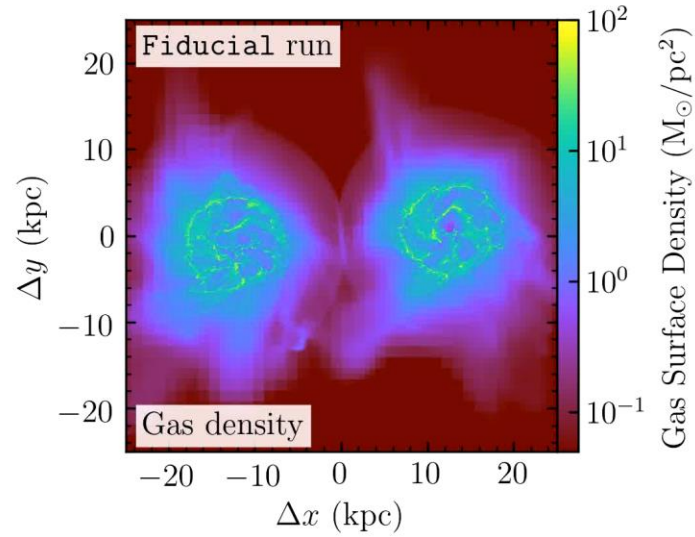


$\times 7$ feedback energy

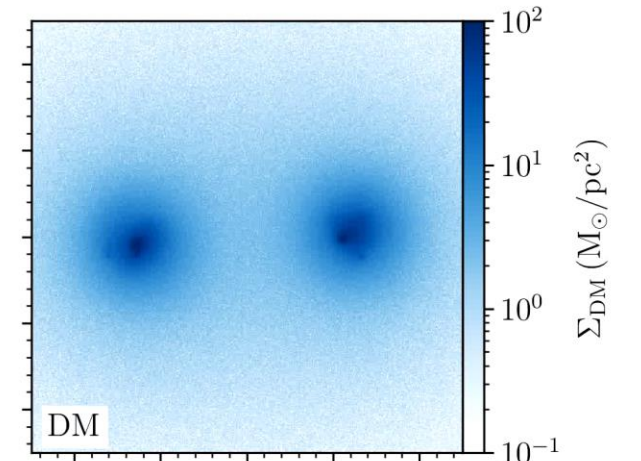
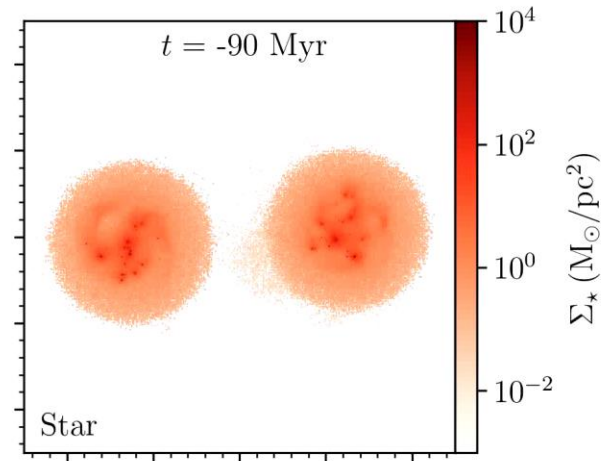
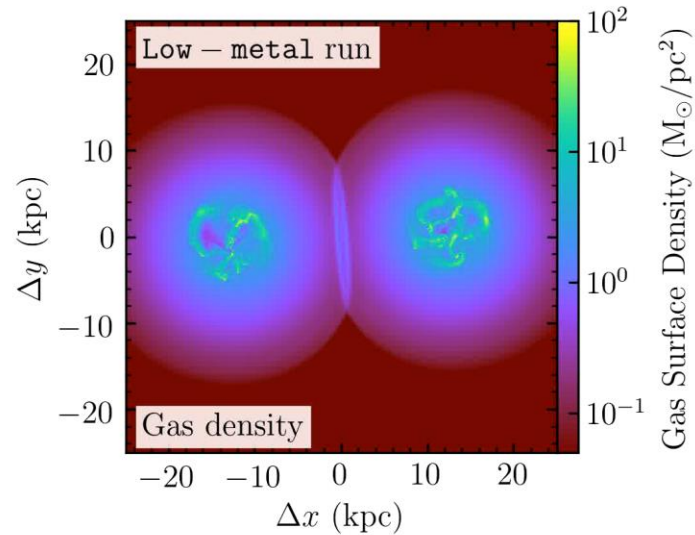


Parameter Study: Initial Metallicity

Fiducial run



Low-metal run
($Z = 0.01 Z_\odot$)



Future work

- Dependence of DMDG and SC formation on initial metallicity needs to be investigated.
 - Stellar feedback (early feedback, radiation feedback, kinetic feedback) is not fully included.
 - Full physics of ISM is not implemented: radiation transfer, chemical chain, photoionization by UV radiation.
- Taking account of these physics, observed metallicity of DMDG and SCs would be explained.
- Numerical resolution (1.25 pc, 400 $M_{\odot}$) only marginally resolves internal structure (half-mass radius, ellipticity) of SCs.
- Improve spatial resolution and resolve individual stars.
- Estimation for probability of galaxy collision in cosmological context.

Take-home Message

- We firstly demonstrated that a high-velocity collision (~ 300 km/s) of gas-rich dwarf galaxies can induce simultaneous formation of dark matter deficient galaxies (DMDGs) with $M_{\text{star}} \sim 10^8 M_{\odot}$ and massive population of star clusters (SCs) with $M_{\text{star}} \sim 10^5 - 10^6 M_{\odot}$.
- Our results from idealized simulation are in accordance with the observed properties of NGC1052-DF2 (and its member globular clusters) and NGC1052-DF4.